Lawrence Du, Ph.D.

Machine Learning Engineering · Computational Biology · MLOps
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Machine learning engineer and biology Ph.D. with experience building scientific AI systems, distributed training platforms, and production models. Work spans sequence foundation models, computer vision, LLM applications, genomics, and experimental molecular biology.

Skills

Languages
Python, SQL, C/C++, C#, Rust, Java, Dart, Bash, HTML/CSS, MATLAB

ML and data tools
PyTorch, TensorFlow, JAX, scikit-learn (sklearn), NumPy, SciPy, pandas, Numba, Ray, MLflow, Flask, Jupyter

Cloud and MLOps
Amazon Web Services (AWS: EC2, S3, Batch, Glue, Athena, Fargate), Google Cloud Platform (GCP), Kubernetes, Docker, Terraform, Pulumi, Metaflow, Jenkins, Linux

ML and scientific computing
Foundation model fine-tuning, deep learning, transformers, convolutional neural networks (CNNs), graph neural networks (GNNs), generative adversarial networks (GANs), diffusion, Louvain/Leiden community detection, SVM, PCA, KNN, decision trees, cross-validation

Experience

GSK · Senior AI/ML Engineer, Sequence Learning and Phenomics
Oct 2024 – Present | South San Francisco, CA

- Built an end-to-end fine-tuning platform for sequence foundation models, enabling internal teams to predict RNA properties from nucleotide sequences and perturbation responses from single-cell RNA-seq. Cut iteration time **from weeks to less than a day**.
- Integrated Segment Anything Model (SAM) image segmentation into production computer vision and image-embedding pipelines, enabling single-cell tracking and cell-state world modeling across **millions of images**.
- Designed embedding-based differentiation quality control (QC) that scored **>3 million iPSC-to-microglia cells**, supporting active Alzheimer's disease research.
- Piloted generative AI using x-prediction / JiT diffusion models to generate protein subcellular localization images conditioned on ESM protein embeddings.
- Built a pipeline using large language models (LLMs) for retrieval-augmented generation (RAG) and gene-candidate evaluation. Prioritized **~1,000 candidate genes** for a lung-epithelium CRISPR knockout screen supporting a next-generation COPD drug program.
- Shipped Model Context Protocol (MCP) servers that let company-wide LLM assistants build and launch scientists' image-analysis pipelines, making internal AI Science tools agent-callable and accelerating analysis turnaround **>10×**.
- Contributing to LLM benchmarking currently in development; conducted interviews to support hiring on the team.

Freenome · Senior Machine Learning Research Engineer
Aug 2022 – Jun 2024 | South San Francisco, CA

- Led a greenfield distributed ML platform project using PyTorch, Ray, and Kubernetes for cancer detection from methylated DNA and protein data. Enabled larger models and accelerated model training **>10×** using distributed data parallelism (DDP).
- Deployed organization-wide MLflow experiment tracking on GCP using Terraform and Pulumi. Enabled live training monitoring, results sharing, and automated reproducible reports, reducing manual researcher effort **≥5×**.
- Built scalable multitask neural networks and elastic net models in PyTorch for colorectal cancer risk from cell-free DNA, supporting a clinical trial cohort of **>27,000 individuals**.
- Piloted biomedical natural language processing (NLP) with GPT-4 and an open-source LLM fine-tuned using direct preference optimization (DPO). Demonstrated literature summarization and feature extraction from unstructured biomedical records.

23andMe · Machine Learning Engineer
Apr 2020 – Aug 2022 | Sunnyvale, CA

- Built large-scale data pipelines for quality statistics and feature engineering of genotyped and imputed single-nucleotide polymorphisms (SNPs). Implemented extract, transform, load (ETL) workflows for **~10 million samples × ~1 million SNPs** using AWS Batch, Metaflow, and Glue, populating an Athena feature datastore for genome-wide association studies (GWAS) and polygenic risk scores (PRS).
- Improved type 2 diabetes and coronary artery disease PRS models through stacking ensembles in production pipelines. Improved sensitivity and specificity for **tens of thousands of customers**, evaluating models using area under the ROC curve (AUC) and precision-recall curve (auPRC) metrics.
- Automated model evaluation and classifier reports with MLflow artifact storage and headless Jupyter execution, reducing researcher analysis time **from days to minutes**.

23andMe · Data Scientist, Ancestry Product
Nov 2018 – Apr 2020 | Sunnyvale, CA

- Created and deployed Recent Ancestor Locations (RAL), a high-precision, high-recall country-matching algorithm serving **>15 million customers** worldwide. Deployed the model as an AWS Fargate microservice with MLflow artifact tracking.
- Improved graph-based techniques for identifying genetic populations from identity-by-descent (IBD) family relationships. Used graph community detection to identify subpopulations in Mexico and the United Kingdom.

Scripps Research · Bioinformatician IV
May 2018 – Oct 2018 | San Diego, CA

- Developed an organ transplant rejection classifier from RNA sequencing data for a research grant.
- Built automated Oxford Nanopore long-read bioinformatics pipelines in Common Workflow Language (CWL) for real-time microbial diagnostics,*de novo* genome assembly, and variant calling.

Juno Diagnostics · Independent Consultant

Sep 2017 – Feb 2018 | San Diego, CA

- Developed a TensorFlow deep learning software package for detecting prenatal genetic abnormalities from high-throughput sequencing data; co-invented a patent for deep learning-based prenatal testing.
- Created a genetic-abnormality simulator derived from statistical analysis of human high-throughput sequencing data.

Insight · Data Science Fellow

Jan 2017 – Apr 2017 | Remote / San Diego, CA

- Built DeepPixelMonster, a TensorFlow deep convolutional generative adversarial network (DCGAN) for pixel art, and deployed it as an interactive Flask application on AWS EC2.

UC San Diego · Ph.D. Researcher, Biology · Scott A. Rifkin Lab

Aug 2010 – May 2017 | La Jolla, CA

- Built DeepNuc, a CNN for classifying **>500,000** transcription start site (TSS) flanking sequences from humans, mice, fruit flies, and nematodes, plus **>60,000** microRNA target sequences from publicly available CLEAR-CLIP data.
- Studied developmental RNA expression noise in **>5,000 embryos** using single-molecule imaging and custom MATLAB tools for image segmentation, fluorescence quantification, and deconvolution.
- Generated **>70 transgenic nematode and Drosophila strains** using Gibson assembly, CRISPR/Cas9, MosSCI, and PhiC31 integrase.

Education

Ph.D., Biology · UC San Diego
2010–2017 • Scott A. Rifkin Lab

B.A., Biological Sciences · Cornell University
2006–2010 • Genetics and Development • *magna cum laude*

Selected Publication and Patent

Du et al. (2016). Mutagenesis of GATA motifs controlling the endoderm regulator elt-2 reveals distinct dominant and secondary cis-regulatory elements. *Developmental Biology*, 412(1), 160–170.

Co-inventor: US20210020314A1: Deep learning-based prenatal testing.

Selected Projects

Rogue Stargun · Unity / C# VR space simulator

- Built spaceship controllers, collision avoidance and firing prediction algorithms, and behavior trees for AI flight planning and decision making.
- Integrated text-to-speech services from Google Cloud, Microsoft Azure, and ElevenLabs into the Unity editor to prototype campaign dialogue; built procedural mission and powerup generation.
- Optimized shaders and rendering/batch calls for constrained VR hardware, achieving **72 FPS on Meta Quest 2 and 90 FPS on Meta Quest 3**

DiffuMon · Open-source image generation

- Implemented denoising diffusion probabilistic models (DDPM) in PyTorch as an educational paper reproduction, with denoising diffusion implicit model (DDIM) sampling support.
- Provided pretrained Fashion MNIST and Pokémon image models, training and sampling commands, configurable datasets, and example notebooks. Designed the implementation to train on a commodity GPU.